Correct
Mark 1.00 out of 1.00

Question text

How many subdirectories will be counted by the following program in the directory "`/tmp/TEST_DIR`", also created by that code? It is supposed that the "`/tmp/TEST_DIR`" **does not exist before the program is run**, and both its creation and those of its subdirectories succeed.

```
int main(int argc, char **argv)
{
    DIR *dir_handle;
    struct dirent *dir_elem;
    struct stat inode;
    char path[1024];
    int dir_count = 0;

    mkdir("/tmp/TEST_DIR");

    for (int i = 0; i < 388; i++) {
        snprintf(path, 1024, "/tmp/TEST_DIR/DIR%03d", i);
        mkdir(path);
    }

    dir_handle = opendir("/tmp/TEST_DIR");

    while ((dir_elem = readdir(dir_handle)) != NULL) {
        snprintf(path, 1024, "/tmp/TEST_DIR/%s", dir_elem->d_name);
        lstat(path, &inode);
        if (S_ISDIR(inode.st_mode))
            dir_count++;
    }

    printf("dir_count = %d\n", dir_count);
}
```

Answer: 390

Question 2

Correct
Mark 1.00 out of 1.00

Question text

Which is the **maximum number of text lines** the following code could read from a text file, supposing the file exists, could be opened for reading, but its length and contents could be variable (i.e. not known in advanced), though the file's length is always greater than the value of "MAX"?

```
#define MAX 833

char lines[MAX];

int fd = open("TEST.txt", O_RDONLY);
if (fd < 0) {
    perror("Cannot open file");
    exit(1);
}

int n = read(fd, lines, MAX);

int no_of_lines = 0;
for(int i = 0; i < n; i++) {
    if (lines[i] == '\n') {
        no_of_lines++;
    }
}

printf("no of lines = %d\n", no_of_lines);
```

Answer: 833

Question 3

Correct
Mark 1.00 out of 1.00

Question text

How many page table entries are there in a page table of a process on a system using 51 bits for memory addresses and pages of 2^{17} bytes?

NOTE: You must introduce the power of 2 of the computed number, i.e. if it is of the form 2^{10}, your response should be 10.

Answer: 34

Which command does the following code, supposed to be executed be a shell, handle?

```
int fd[2];

pipe(fd);

pid = fork ();
if ( pid > 0) {
    pid = fork ();
    if ( pid > 0) {
        close(fd[0]);
        close(fd[1]);
        wait(NULL);
        wait(NULL);
    } else {
        dup2(fd[0], 0);
        close(fd[0]);
        close[fd[1]);
        execlp("cat", "cat", NULL);
        perror("execl has not succeded");
    }
} else {
    dup2(fd[1], 1);
    close(fd[0]);
    close(fd[1]);
    execlp("cat", "cat", "mouse", NULL);
    perror("execl has not succeded");
}
```

Select one:

- a. cat mouse cat
- b. cat mouse | cat